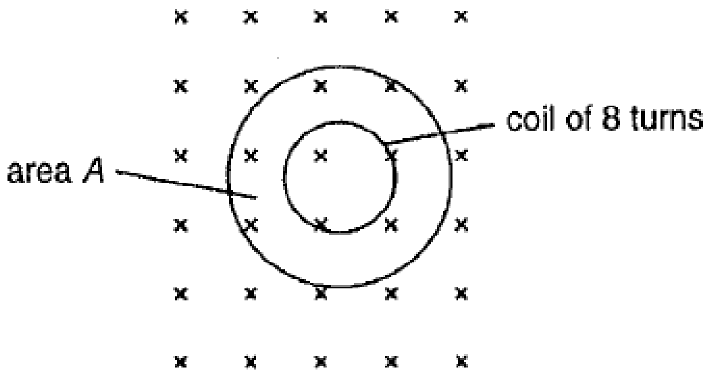
23	A uniform magnetic field of flux density B passes normally through a plane area A. In this plane lies a coil of eight turns of wire, each of area ¼ A.  What is the magnetic flux linkage for the coil in terms of B and A?

A	¼ BA	**B**	$2BA$
C	$4BA$	**D**	$8BA$

Ans: B

$$\varphi = NBA = 8B\left(\frac{1}{4}A\right) = 2BA$$

24	A flat, circular coil of wire 30 turns, each of area 0.025 m², is initially placed with its plane